SCHOOL MANAGEMENT SYSTEM

Java OOP Project Report

School Management System

A Java OOP-Based Application with SQLite Integration

- **Programming Language:** Java
- **Database:** SQLite (via JDBC)
- **Paradigm:** Object-Oriented Programming
- **IDE:** Visual Studio Code
- **Access Roles:** Principal · Teacher · Student
- **Academic Session:** 2024–25

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1. Abstract

The School Management System is a Java-based desktop application designed to digitize and streamline administrative operations of educational institutions. The system is built using Object-Oriented Programming (OOP) principles and implements role-based access control for three types of users: Principal, Teacher, and Student.

The application uses SQLite as an embedded database connected via JDBC, eliminating the need for an external server. The system supports full CRUD operations and demonstrates modular design, encapsulation, and scalability suitable for academic and real-world applications.

2. Introduction

Educational institutions handle large volumes of data such as student records, teacher details, and academic information. Manual systems are inefficient and prone to errors. Therefore, a digital solution is necessary.

This project provides a structured and role-based system developed in Java. It ensures secure data handling through encapsulation and maintains persistent storage using a lightweight database. The project also serves as a practical implementation of OOP concepts.

3. Objectives

- To develop a school management system using Java
- To implement core OOP principles
- To provide role-based access control
- To integrate SQLite using JDBC
- To support CRUD operations
- To create a modular and maintainable system

4. Scope of the Project

The system is designed for small to medium-sized institutions and includes:

- Role-based authentication system
- Management of student, teacher, and subject data
- Database storage using SQLite
- Command-line interface for system interaction

Advanced features like GUI, cloud support, and analytics are not included and are reserved for future enhancements.

5. Methodology

Phase	Activity	Output
Requirement Analysis	Identified roles and system needs	Feature specification
System Design	Designed classes and database schema	UML & ER diagrams
Implementation	Developed Java code and DB integration	Working system
Testing	Verified functionality and access control	Test results
Documentation	Prepared report and documentation	Final report

6. Technologies Used

- **Programming Language:** Java (JDK 17+)
- **IDE:** Visual Studio Code
- **Database:** SQLite
- **Connectivity:** JDBC
- **Runtime:** JVM
- **Paradigm:** OOP (Encapsulation, Abstraction, Inheritance, Polymorphism)

7. System Architecture & OOP Design

7.1 Class Structure

- Base class: `User`
- Derived classes: `Principal`, `Teacher`, `Student`
- Database handler: `DBManager`

7.2 OOP Principles Used

- **Encapsulation:** Data hidden within classes
- **Abstraction:** Common interface through base class
- **Inheritance:** Roles extend User class
- **Polymorphism:** Methods overridden based on roles

8. User Roles and Functionalities

Principal

- Full system access
- Can manage all records
- Can perform CRUD operations

Teacher

- Limited access
- Can manage subject-related data

Student

- Read-only access
- Can view personal and academic details

9. System Features

- User authentication
- Role-based authorization
- CRUD operations
- SQLite database integration
- Input validation
- Modular code structure

10. Database Design

The system uses SQLite, a file-based database.

Main Tables:

- **Users:** Authentication and role data
- **Students:** Student information
- **Teachers:** Teacher data
- **Subjects:** Subject details

SQLite is chosen because it is lightweight, portable, and requires no server setup.

11. Advantages

- No server required (SQLite-based)
- Lightweight and portable
- Strong demonstration of OOP concepts
- Role-based security
- Easy to maintain and extend

12. Limitations

- No graphical interface
- Limited scalability
- Basic security features
- No real-time analytics
- Single-system usage (no network support)

13. Future Enhancements

- GUI using JavaFX or Swing
- Web-based version (Spring Boot + React)
- Attendance management system
- Fee management module
- Advanced authentication
- Multi-user support using MySQL/PostgreSQL
- Performance analytics dashboard

14. Conclusion

The School Management System successfully demonstrates the implementation of Java OOP concepts with database integration. It provides a structured, modular, and role-based system for managing school data efficiently.

The project is scalable and can be extended with advanced features such as GUI and web-based architecture, making it a strong foundation for real-world applications.

15. References

1. Oracle Java Documentation
2. SQLite Documentation
3. JDBC API Guide
4. Effective Java – Joshua Bloch
5. Core Java – Cay S. Horstmann
6. Academic course materials